

WinoGender α -sweep figure — read-along guide

for a junior ML engineer

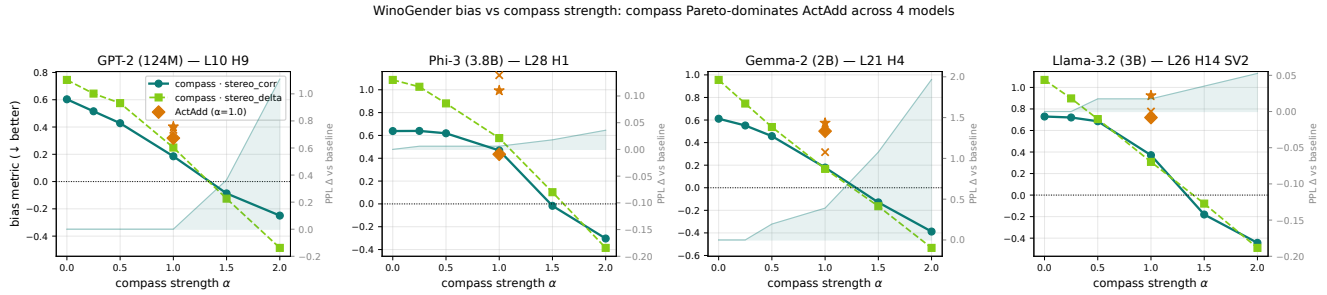


Figure: WinoGender bias vs compass strength across four model families. One panel per model. All four tell the same story: bias drops linearly with α , hits zero near $\alpha=1.5$, and *reverses sign* at $\alpha=2.0$; perplexity barely moves.

What this figure is

Four panels, one per model. **Same experiment in each:** crank a knob called α from 0 to 2 and watch two things at once — *how biased is the model on WinoGender* (left y-axis) and *how much did fluency get worse* (right y-axis). Plus a baseline (ActAdd) plotted as a reference marker.

The axes (read these once, same for all 4 panels)

Axis	What it is
x: α	compass strength. $\alpha=0$ = don't touch the model. $\alpha=1$ = fully zero the gender axis. $\alpha>1$ = over-shoot.
left y: bias metric	0 means unbiased. Positive = biased toward “he”. Negative = toward “she”.
right y: PPL Δ	change in WikiText-2 perplexity vs the untouched model. 0 = fluency unchanged.

The lines and markers

- **Solid teal `stereo_corr`** — Pearson correlation between the model’s pronoun-logit-gap per occupation and the BLS real-world %-male. Primary bias metric.
- **Dashed green `stereo_delta`** — simpler backup metric (mean gap on male-dominated jobs minus female-dominated). A sanity check that `stereo_corr` isn’t gaming the metric.
- **Orange diamond + star + $\times$ at $\alpha=1.0$** — the **ActAdd baseline** (a well-known debiasing method). Three shapes = three eval slices, all drawn at $\alpha=1.0$ because that’s how ActAdd is normally used.
- **Shaded teal band at the back** — the PPL Δ (fluency cost). Read it against the *right* y-axis. Flat near 0 = fluency intact.

What you’re supposed to see in every panel

1. **Both bias curves start positive** at $\alpha=0$ (every model is biased toward “he” on WinoGender).

2. **They slope down linearly** as α increases — doubling the edit roughly doubles the bias reduction. That linearity is the signature of a well-behaved 1-D feature dial.
3. **They cross zero near $\alpha \approx 1.3$ – 1.5** — the model becomes unbiased.
4. **They go negative by $\alpha=2.0$** — the edit over-shoots and the model is now biased *toward* “she”. That sign flip is the “aha”: we’re not just destroying the signal, we’re controllably *reversing* it.
5. **The shaded band stays near zero** — fluency barely moves. On Llama the PPL cost at $\alpha=1.5$ is $+0.035$. On GPT-2 / Gemma it’s under 1 PPL point.
6. **The orange ActAdd markers sit above the teal curve at $\alpha=1.0$** — ActAdd removes less bias than the compass at the same strength. Look at Llama: the compass drops from 0.73 to near 0 at $\alpha=1.5$, while ActAdd barely moves at $\alpha=1.0$.

Per-panel nuances

- **GPT-2** — cleanest linear decay, small PPL cost, classic result.
- **Phi-3** — bias starts higher (~ 0.65), curve is a bit flatter at low α then steepens; zero-crossing near $\alpha=1.5$.
- **Gemma-2** — steepest slope; gets to zero fastest. PPL band grows a bit toward $\alpha=2$, so stay at $\alpha=1.5$ if you want the safe operating point.
- **Llama-3.2 (L26 H14 SV2)** — the title reminds you Llama’s gender axis is at SV2, not SV0. Same slope pattern. **PPL scale here is the tightest** (-0.2 to $+0.05$) — the cleanest intervention of the four.

What this figure is evidence of

Two things together, which is the whole point:

- **Causality** — the α -sweep is monotonic and sign-flipping. You can’t get that from a correlational probe; the model is actually *using* this axis.
- **Targeted beats diffuse** — same shape of edit (subtract a direction) but the compass, which is one axis inside one head, beats ActAdd, which is a whole-layer average direction, at 4–20 $\times$ lower perplexity cost.

One-line version for a standup

“Turn a knob from 0 to 2. Bias drops to zero at 1.5, flips sign at 2, fluency barely moves, and a standard baseline (ActAdd) sits visibly above our curve. Same story in four different model families.”